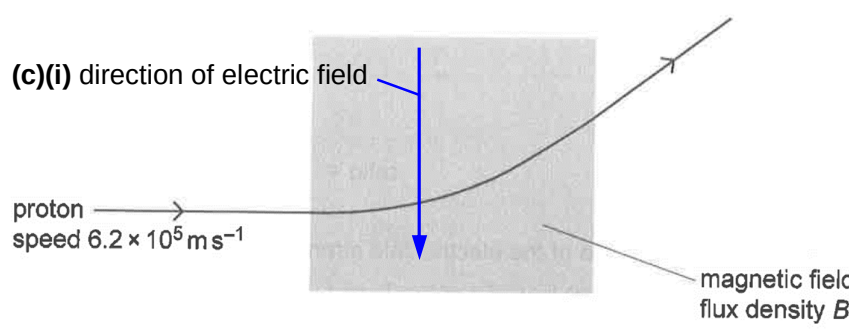
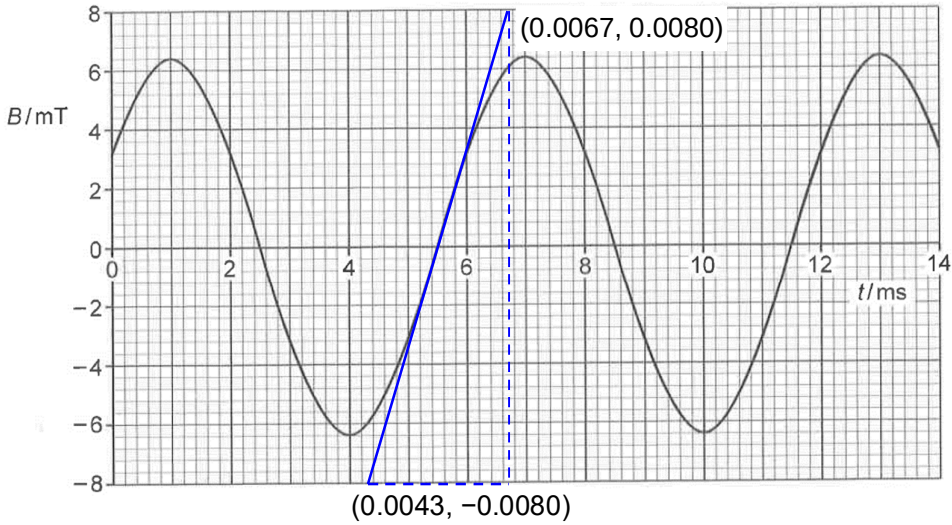


	$\approx 2.14 \times 10^{-10} \text{ N/C}$
8(a)	Magnetic flux density is defined as the force per unit length per unit current acting on a straight conductor placed perpendicularly to the magnetic field.
8(b)(i)	As the proton enters the magnetic field at right angles to the magnetic field, it experiences a magnetic force that is at right angles to both the magnetic field and its velocity. The magnetic force does not change the magnitude of its velocity but will constantly change its direction. The magnetic force provides the centripetal force for the motion of the proton along the arc of a circular path.
8(b)(ii)	Magnetic force on the proton provides for its centripetal force $F_B = \frac{mv^2}{r}$ $Bev = \frac{m_p v^2}{r}$ $B = \frac{m_p v}{er}$ $= \frac{(1.67 \times 10^{-27})(6.2 \times 10^5)}{(1.60 \times 10^{-19})(7.6 \times 10^{-2})}$ $= 0.08515$ $\approx 0.0852 \text{ T}$
8(c)(i)	 (c)(i) direction of electric field proton speed $6.2 \times 10^5 \text{ m s}^{-1}$ magnetic field flux density B Fig. 8.1 (not to scale)
8(c)(ii)	For undeviated path, the magnitudes of the magnetic force and electric force must be equal.

No.	Solution
	$F_B = F_E$ $Bev = eE$ $E = Bv$ $= 0.08515 \times 6.2 \times 10^5$ $= 5.2793 \times 10^4$ $\approx 5.28 \times 10^4 \text{ V m}^{-1}$
8(d)(i)	r.m.s. value of the magnetic flux density in coil S $= \frac{B_0}{\sqrt{2}}$ $= \frac{6.4 \times 10^{-3}}{\sqrt{2}}$ $= 4.525 \times 10^{-3}$ $\approx 4.53 \text{ mT}$
8(d)(ii)	Two times at which the e.m.f. induced in coil S is zero. 1.0 ms, 4.0 ms Other times at which the e.m.f. induced in coil S is zero. 7.0 ms, 10.0 ms, 13.0 ms
8(e)	From Fig. 8.3, $B_0 = 6.4 \text{ mT}$, 1.5 cycles takes $(11.5 - 2.5) = 9.0 \text{ ms}$ So period, $T = 6.0 \text{ ms}$ Since magnetic flux density varies sinusoidally, $B = B_0 \sin(\omega t)$, By Lenz's law, $E = \frac{d(NBA)}{dt}$ $= NA \frac{dB}{dt}$ $= NAB_0 \omega \cos(\omega t)$ Maximum e.m.f. induced in coil S $E_{\max} = NAB_0 \omega$ $= (270) \left(\frac{\pi}{4} \times (2.4 \times 10^{-2})^2 \right) (6.4 \times 10^{-3}) \left(\frac{2\pi}{6.0 \times 10^{-3}} \right)$ $= 0.8186$ $\approx 0.819 \text{ V}$

No.	Solution
	Alternatively, using graphical approach,  Fig. 8.3 Taking tangent at a point where the graph has maximum gradient, the gradient coordinates are (0.0066, 0.0080) and (0.0043, -0.0080). Maximum e.m.f. induced in coil S, $ \begin{aligned} E_{\max} &= \left(\frac{d\Phi}{dt} \right)_{\max} \\ &= \left(\frac{d(NBA)}{dt} \right)_{\max} \\ &= NA \left(\frac{dB}{dt} \right)_{\max} \\ &= (270) \left(\frac{\pi}{4} \times (2.4 \times 10^{-2})^2 \right) \left(\frac{0.0080 - (-0.0080)}{0.0067 - 0.0043} \right) \\ &= 0.8143 \\ &\approx 0.814 \text{ V} \end{aligned} $ <i>Note:</i> The maximum e.m.f. by calculation and graphical method will not be exactly the same because the tangent draw on Fig. 8.3 is approximated. The accepted range is 0.78 – 0.86 V.

No.	Solution
8(f)	Fig. 8.5 Using Faraday's law of electromagnetic induction, $E \propto -\frac{d(N\phi)}{dt}$ <i>Note: The common mistake is not realising that the e.m.f. is positive when the change in flux is negative.</i>
9(a)(i)	Two pieces of evidence: